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Cambridge IGCSE Combined and Co-ordinated Sciences

B2 Movement into and out of Cells

Condensed Version — Key Concepts and High-Probability Exam Questions

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Cambridge IGCSE Biology · The most important concepts and exam questions for your test

Must-Know Concepts

Diffusion

- **Diffusion** is the net movement of particles from a region of their higher concentration to a region of their lower concentration (i.e. down a concentration gradient), as a result of their random movement.
- **Net movement** means overall movement: particles move **randomly**, but more leave the high concentration than enter it.
- Diffusion is **passive**: its energy is the **kinetic energy** of the particles, not energy from respiration.
- **Oxygen** and **carbon dioxide** diffuse through the **cell membrane**. A respiring cell uses up oxygen, so oxygen diffuses in; it makes carbon dioxide, which diffuses out.

Factors Affecting the Rate of Diffusion

Factor	Faster diffusion with	Why
Temperature	A higher temperature	More kinetic energy, so quicker random movement
Surface area	A larger surface area	A greater area over which particles can move
Concentration gradient	A steeper gradient	More particles move towards the low concentration than move back, so the net movement is greater
Distance	A shorter distance, e.g. a thin membrane	Less distance to travel

Osmosis and Water Potential

- **Osmosis** is the diffusion of water molecules through a partially permeable membrane.
- **In terms of water potential**, osmosis is the net movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a partially permeable membrane.
- **Cell membranes are partially permeable**; the plant **cell wall is fully permeable**.
- **Dilute** = a lot of water = **high water potential**. **Concentrated** = less water = **low water potential**. **Pure water** has the highest water potential.

Osmosis in Animal and Plant Cells

Cell	In pure water	In a concentrated solution
Animal cell	Water enters by osmosis; the cell swells and **bursts**, as it has no cell wall	Water leaves by osmosis; the cell shrinks
Plant cell	Water enters; the cytoplasm and vacuole swell and press on the cell wall; the cell becomes **turgid** but does not burst	Water leaves; the cytoplasm and vacuole shrink; the cell becomes **flaccid** and may become **plasmolysed**

- The **strong cell wall** resists the swelling contents and presses back, so a plant cell does not burst. The pressure of the water pushing outwards on the wall is **turgor pressure**; it helps to support

leaves.
- Flaccid cells make a plant **wilt**. **Plasmolysis** can kill a cell, because the cell membrane is damaged as it tears away from the cell wall.

Active Transport

- **Active transport** is the movement of molecules or ions through a cell membrane from a region of lower concentration to a region of higher concentration (i.e. against a concentration gradient) using energy from respiration.
- The energy comes from **aerobic respiration** in the **mitochondria**, so anything that slows respiration (e.g. a shortage of oxygen) slows active transport.
- **Root hair cells** take in **nitrate ions** this way, even though the concentration of nitrate ions inside the cell is usually higher than in the soil.

How a **carrier protein** in the cell membrane works:

1. The nitrate ion enters the carrier protein from outside the cell.
2. The carrier protein changes shape, using energy from respiration.
3. The change of shape pushes the nitrate ion into the cell.

Diffusion, Osmosis and Active Transport Compared

Feature	Diffusion	Osmosis	Active transport
What moves	Any particles, e.g. O_2, CO_2	Water molecules only	Molecules or ions, e.g. nitrate
Direction	High → low concentration	High → low water potential	Low → high concentration
Energy	Kinetic energy of particles	Kinetic energy of water molecules	Respiration
Carrier proteins	No	No	Yes

Key Terms

Term	Definition
Particles	(In this context) the smallest pieces of which a substance is made; particles can be molecules, atoms or ions
Diffusion	The net movement of particles from a region of their higher concentration to a region of their lower concentration (i.e. down a concentration gradient), as a result of their random movement
Net movement	Overall or average movement
Concentration gradient	An imaginary 'slope' from a high concentration to a low concentration
Independent variable	The variable that you change in an experiment
Dependent variable	The variable that you measure as you collect your results
Partially permeable membrane	A membrane that lets some molecules through (e.g. small water molecules) but not others (e.g. much larger sugar molecules)

Term	Definition
Fully permeable	Lets all molecules pass through; the cell wall of a plant cell is fully permeable
Osmosis	The diffusion of water molecules through a partially permeable membrane
High water potential	An area where there are a lot of water molecules, that is, a dilute solution
Low water potential	An area where there are not many water molecules, that is, a concentrated solution
Water potential gradient	A difference in water potential between two areas
Turgid	A description of a plant cell that is tight and firm
Turgor pressure	The pressure of the water pushing outwards on a plant cell wall
Flaccid	A description of a plant cell that has lost its turgor pressure and is soft
Plasmolysed	A description of a cell in which the cell membrane tears away from the cell wall
Active transport	The movement of molecules or ions through a cell membrane from a region of lower concentration to a region of higher concentration (i.e. against a concentration gradient) using energy from respiration
Carrier proteins (protein carriers)	Protein molecules in cell membranes that can use energy to change shape and move ions or molecules into or out of a cell

Key Calculations and Experiments

Surface Area and Volume

- **Volume of a cube** = side × side × side (cm^3). **Surface area of a cube** = 6 × side × side (cm^2).
- Cutting a cube into smaller cubes increases the total surface area but not the total volume, so substances diffuse in faster.

Agar Jelly Experiment (Diffusion)

- Alkaline agar jelly with **cresol red** indicator is **red-purple** and turns **yellow** as acid diffuses in. A fully yellow cube means the acid has reached its centre.

Potato Strip Experiment (Osmosis)

- Strips of equal length (e.g. 40 mm) are left in distilled water, a dilute sugar solution and a concentrated sugar solution for about 20 minutes, then measured again.
- **Change in length** = final length − starting length (positive = longer, negative = shorter).
- **Percentage change** = (change in length ÷ starting length) × 100%.
- **Distilled water**: longer and firm (cells turgid). **Concentrated sugar**: shorter and soft (cells flaccid). **Dilute sugar**: a small change either way, depending on whether its water potential is higher or lower than that of the cells.

Planning an Investigation

- Change one **independent variable**, measure the **dependent variable** and keep the **standardised variables** the same so they cannot affect the results. Use at least five values, repeat and calculate a **mean**, and put units only in the column headings (e.g. Temperature / °C).
- An **anomalous result** does not fit the trend: pick the point furthest from the line of best fit and say why.

High-Probability Exam Questions

Question 1

Which change would increase the rate of diffusion across a membrane? **A** a decrease in concentration gradient; **B** a decrease in surface area; **C** an increase in distance across the membrane; **D** an increase in temperature. (1 mark)

Answer:

- **D**: particles have more kinetic energy, so their random movement is quicker. A, B and C would all slow diffusion down.

Question 2

A cube with 1 cm sides and eight cubes with 0.5 cm sides, all cut from alkaline agar jelly containing cresol red, are covered with dilute hydrochloric acid.

a) Calculate the total volume and the total surface area of the eight small cubes, and compare each with the large cube. (4 marks)

b) Copy and complete these sentences, choosing the correct words: "The time taken for the eight small cubes to completely change colour was longer/shorter than the time for the one single cube. This is because the surface area/volume of the small cubes was greater/smaller than that of the single cube." (3 marks)

(Total: 7 marks)

Answer:

- **a)** Small cubes: total volume = 8 × (0.5 × 0.5 × 0.5) = 8 × 0.125 = 1 cm³.
- **a)** Large cube: 1 × 1 × 1 = 1 cm³, so the total volume is the same.
- **a)** Small cubes: total surface area = 8 × 6 × (0.5 × 0.5) = 8 × 1.5 = 12 cm².
- **a)** Large cube: 6 × (1 × 1) = 6 cm², so the small cubes have twice the surface area.
- **b)** Shorter.
- **b)** Surface area.
- **b)** Greater: 12 cm² compared with 6 cm², so the acid can diffuse in across a larger area (and has a shorter distance to travel to the centre of each cube).

Question 3

You plan to use the same agar cubes and dilute hydrochloric acid to investigate how temperature affects the rate of diffusion.

a) The variable that you change in your experiment is called the independent variable. What is the independent variable in your experiment? (1 mark)

b) The variable that you measure, as you collect your results, is the dependent variable. What is the dependent variable in your experiment? (1 mark)

(Total: 2 marks)

Answer:

- **a)** The temperature of the hydrochloric acid that the cubes are placed in, measured in °C.
- **b)** The time taken for an agar cube to turn completely yellow, measured in seconds (the shorter the time, the faster the rate of diffusion).

Question 4

Tubes of red dye were kept in water-baths at seven temperatures for ten minutes. A cube of agar jelly was then put into each tube and the time for it to change colour was measured. The straight line of best fit falls from about 12 minutes at 0 °C to below 1 minute at 60 °C. Values read from the graph:

Temperature / °C	0	10	20	30	40	50	60
Time taken / minutes	12.0	10.6	4.5	6.1	4.0	2.4	1.2

a) Name the process by which the dye moved through the jelly, and describe this process. (3 marks)

b) Suggest why the tubes were left in the water-baths for ten minutes. (1 mark)

c) State two variables that should be standardised. (2 marks)

d) i) One result is anomalous. Identify it and explain your choice. (2 marks)

d) ii) Describe the trend shown by the results. (1 mark)

d) iii) Use your knowledge of diffusion to explain the results. (2 marks)

(Total: 11 marks)

Answer:

- **a)** Diffusion.
- **a)** The net movement of dye particles from a region of higher concentration (the dye solution) to a region of lower concentration (the jelly), down a concentration gradient.
- **a)** As a result of the random movement of the particles (from their kinetic energy).
- **b)** So that the dye reached the temperature of its water-bath before the cube was added.
- **c)** The size of the jelly cubes.
- **c)** The concentration of the dye (or: the volume of dye; jelly from the same batch; the same end-point).
- **d) i)** The result at 20 °C (about 4.5 minutes).
- **d) i)** It is far below the line (which predicts about 8 minutes) and shorter than the time at 30 °C, so it does not fit the trend.
- **d) ii)** As the temperature increases, the time taken for the cubes to change colour decreases.
- **d) iii)** At higher temperatures the dye particles have more kinetic energy, so their random movement is faster.
- **d) iii)** So the dye diffuses into the jelly faster and the cube changes colour in less time.

Question 5

A dilute sugar solution is separated from a concentrated sugar solution by a membrane that lets water molecules through but not sugar molecules. Which solution has the higher water potential? Explain your answer. (2 marks)

Answer:

- The dilute sugar solution.
- It contains more water molecules (less of its space is taken up by sugar), and a solution with a lot of water has a high water potential.

Question 6

a) Describe what is meant by osmosis. (2 marks)

b) Compare the effect of immersing an animal cell and a plant cell in pure water. Explain the reasons for any differences that you describe. (6 marks)

(Total: 8 marks)

Answer:

- **a)** The net movement (diffusion) of water molecules...
- **a)** ...from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a partially permeable membrane.
- **b)** Any six of:
- Both cells take in water by osmosis.

- Pure water has a higher water potential than the cytoplasm, so water moves in down a water potential gradient.
- In both, water enters through the partially permeable cell membrane.
- In both, the contents swell.
- The animal cell keeps swelling until it bursts.
- This is because it has no cell wall: its thin cell membrane stretches until the strain is too much.
- The plant cell does not burst; it becomes turgid (firm).
- This is because its strong cell wall resists: the contents press outwards on it (turgor pressure) and the wall presses back.

Question 7

A plant cell in a concentrated solution has shrunk and its cell membrane has pulled away from the cell wall, leaving a gap (space X). What fills space X? Explain your answer. (2 marks)

Answer:

- The concentrated solution from outside the cell (not air, and not cytoplasm, which stays inside the membrane).
- The cell wall is fully permeable, so the outside solution passes straight through it into the gap.

Question 8

Plant cells that have been plasmolysed in a concentrated solution are placed in pure water. Predict what will happen. Explain your answer, using the term water potential. (5 marks)

Answer:

- The cells take up water by osmosis and become turgid again (if they have not been killed).
- Pure water has a higher water potential than the cell contents, which are concentrated solutions.
- So water moves into the cells down the water potential gradient, through the partially permeable cell membranes.
- The cytoplasm and vacuole swell, the membrane is pushed back against the cell wall and turgor pressure builds up.
- A cell whose membrane was badly damaged as it tore away may have died; its membrane is no longer partially permeable, so it will not become turgid.

Question 9

A bar chart shows potassium and sodium ion concentrations (in arbitrary units) in pond water and in the cells of a plant growing in it:

Ion	Pond water	Plant cells
Potassium	0.5	4.0
Sodium	0.5	about 0.05 (almost zero)

a) Describe the differences between the ion concentrations in the pond water and in the plant cells. (3 marks)

b) Use the data to suggest how the ions move between the pond water and the plant cells. Explain your answer. (2 marks)

c) Describe how this process takes place. (4 marks)

(Total: 9 marks)

Answer:

- **a)** Potassium is much more concentrated in the plant cells (4.0 arbitrary units) than in the pond water (0.5 arbitrary units), about 8 times higher.

- **a)** Sodium is less concentrated in the plant cells (about 0.05) than in the pond water (0.5).
- **a)** In the plant cells there is far more potassium than sodium.
- **b)** Potassium ions are moved into the plant cells by active transport.
- **b)** They move from a lower concentration in the pond water to a higher concentration in the cells, against their concentration gradient; diffusion would move them out.
- **c)** Any four of:
- There are carrier proteins in the cell membrane.
- A potassium ion from the pond water enters (binds to) a carrier protein.
- The carrier protein changes shape.
- The energy for the change of shape comes from respiration in the mitochondria.
- The change of shape pushes the ion through the membrane into the cytoplasm.

Common Mistakes to Avoid

- **Saying particles "want" to move.** They move randomly; more of them happen to leave the high concentration than enter it.
- **Saying diffusion or osmosis uses energy from respiration.** Both use the particles' kinetic energy; only active transport uses respiration.
- **Getting water potential the wrong way round.** Dilute = high, concentrated = low; water moves from high to low. Only water moves in osmosis, never the solute.
- **Calling the cell wall partially permeable.** Only the cell membrane is. The wall is fully permeable, so the gap in a plasmolysed cell fills with outside solution, not air.
- **Mixing up flaccid and plasmolysed.** Flaccid: lost turgor pressure, soft. Plasmolysed: the membrane has torn away from the wall.
- **Incomplete active transport answers.** Give all three ideas: against a concentration gradient, carrier proteins that change shape, energy from respiration.

Last-Minute Checklist

- I can define diffusion, osmosis (both ways) and active transport word for word.
- I can explain how four factors affect the rate of diffusion, and calculate the volume and surface area of cubes.
- I can say which of two solutions has the higher water potential and which way water moves.
- I can explain what happens to animal and plant cells in pure water and in a concentrated solution, using turgid, turgor pressure, flaccid and plasmolysed.
- I can explain potato strip results and calculate change in length and percentage change.
- I can describe how carrier proteins carry out active transport.
- I can compare diffusion, osmosis and active transport.
- I can name the variables in an experiment, spot an anomalous result and describe a trend using data.